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Branching

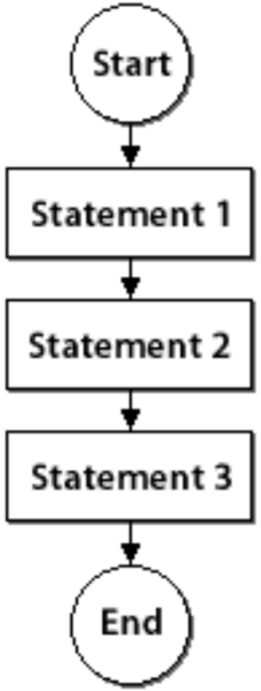
Introduction to Computer Programming

Management and Artificial Intelligence

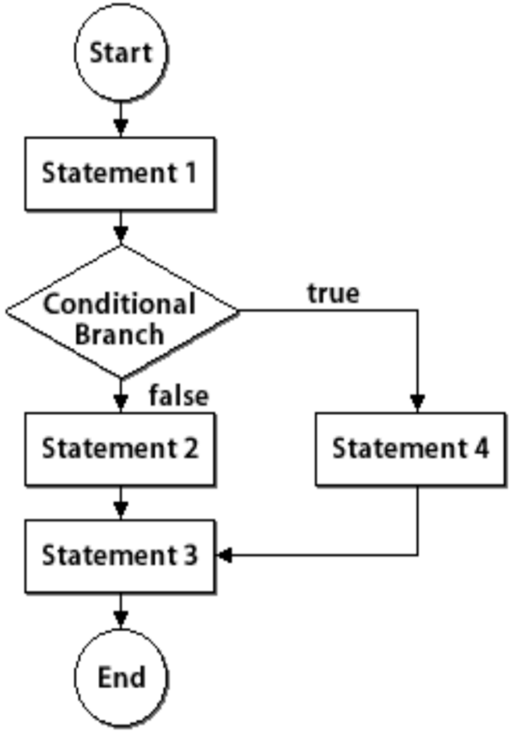
LUISS

Branching

A program **without branching**



A program **with branching**



Branching in Python

Plain if

```
if <condition>:
    statement
    statement
    ...
```

if-else

```
if <condition>:
    statement
    ...
else:
    statement
    ...
```

if-elif-else

```
if <condition>:
    statement
    ...
elif <condition>:
    statement
    ...
else:
    statement
    ...
```

- `<condition>` yields `True` or `False`
- Evaluate statements in that block if `<condition>` is `True`
- Indentation is the key to delimit blocks
- Place `:` at the end of `<condition>`

Examples: if

```python
num = 3
if num > 0:
    print(num, "is a positive number.")
print("This is always printed. It's outside of the block.")
```

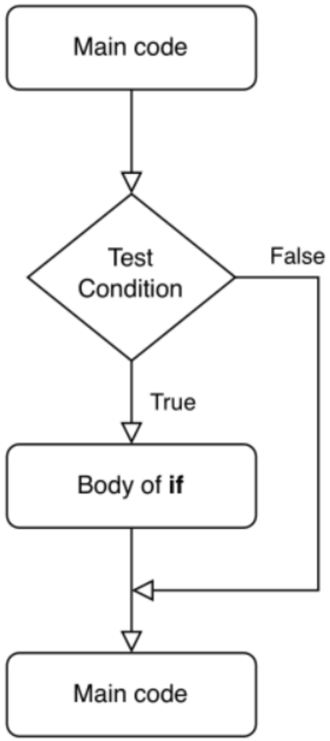
```
3 is a positive number.
This is always printed. It's outside of the block.
```

```python
num = -1
if num > 0:
    print(num, "is a positive number.")
print("This is always printed. It's outside of the block.")
```

```
This is always printed. It's outside of the block.
```

Flowchart of an `if` statement

Logical flowchart of an if statement:



Examples: if-else

```python
num = 1
if num >= 0:
    print("Number is positive or zero.")
else:
    print("Negative number.")
```

```
Number is positive or zero.
```

What happens with:

- `num = 0`
- `num = -5`

?

```python
num = 0
if num >= 0:
    print("Number is positive or zero.")
else:
    print("Negative number.")
```

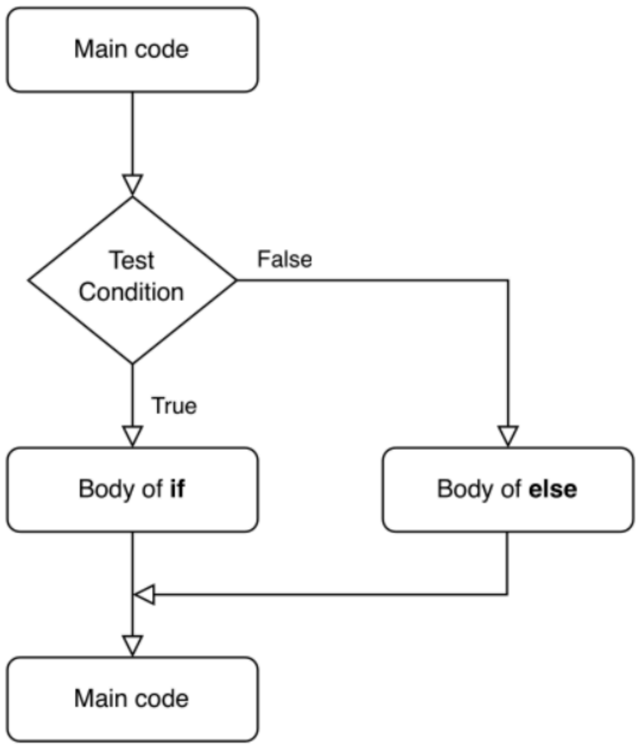
```
Number is positive or zero.
```

```python
num = -5
if num >= 0:
    print("Number is positive or zero.")
else:
    print("Negative number.")
```

```
Negative number.
```

Flowchart of an `if-else` statement

Logical flowchart of an if-else statement.:



Examples: if-elif-else

```python
num = 3.4
if num > 0:
    print("Number is positive.")
elif num == 0:
    print("Number is zero.")
else:
    print("Number is negative.")
```

```
Number is positive.
```

What happens with:

- `num = 0`
- `num = -4.5`

?

```
num = 0
if num > 0:
    print("Number is positive.")
elif num == 0:
    print("Number is zero.")
else:
    print("Number is negative.")
```

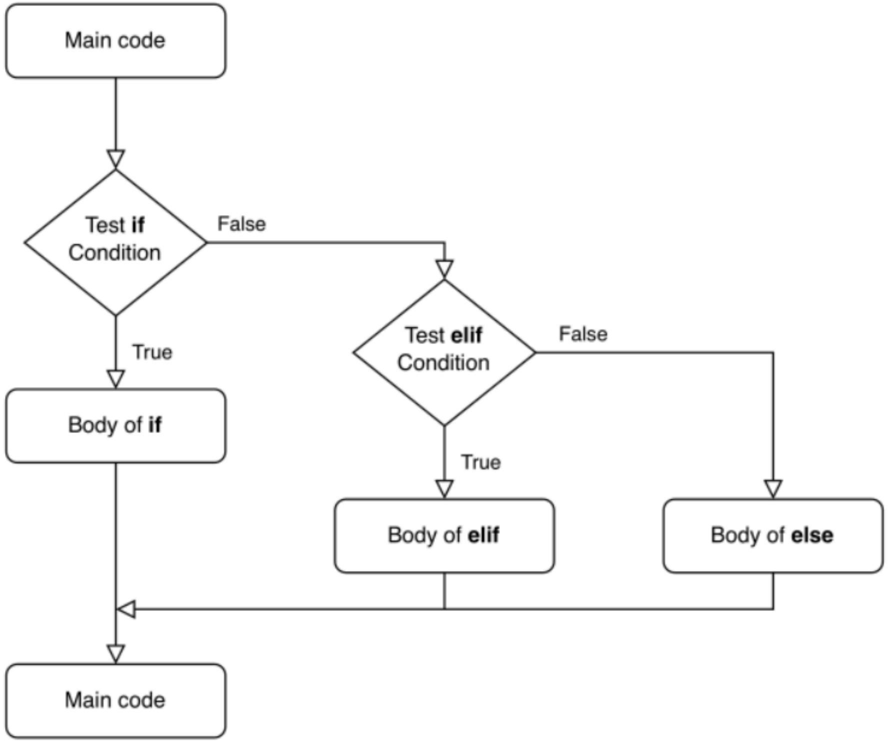
```
Number is zero.
```

```
num = -4.5
if num > 0:
    print("Number is positive.")
elif num == 0:
    print("Number is zero.")
else:
    print("Number is negative.")
```

```
Number is negative.
```

Flowchart of an `if-elif-else` statement

Logical flowchart of an if-elif-else statement:



Nested Statements

Branching statements `if`, `else` and `elif` can be nested arbitrarily.

```
num = 10
# or interactive:
# num = float(input()) # recall: input() returns a string
print("You entered: ", num)
if num >= 0:
    if num == 0:
        print("Number is zero")
    else:
        print("Number is positive")
else:
    print("Number is negative")
```

```
You entered:  10
Number is positive
```

Indentation

Recall: Indentation (tab or 4 spaces) delimits blocks of code in Python.

```python
x = 10.0 #float(input("Enter a value for x: "))
y = 20.0 #float(input("Enter a value for y: "))
print("x is", x)
print("y is", y)
if x == y:
    print("y and x are equal")
    if y != 0:
        print("Finally, x/y is", x / y)
elif x > y:
    print("y is smaller")
else:
    print("x is smaller")
```

```
x is 10.0
y is 20.0
x is smaller
```

Warm-up Exercises

- Notebook with exercises on branching (no hints)
- Notebook with exercises on branching (with pseudocode)
- Notebook with exercises on branching (with solutions)

Exercise Session

Find more exercises here.